REPORT

Learning Word Embeddings from IIT Jodhpur Data

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1. Introduction

In this assignment, we aim to learn meaningful word representations (word embeddings) using textual data collected from IIT Jodhpur sources. Word embeddings help in capturing semantic relationships between words, which are useful in many Natural Language Processing (NLP) tasks. We use two popular models: **CBOW (Continuous Bag of Words)** and **Skip-gram**, both implemented using Word2Vec.

2. Data Collection

The dataset was created from two main sources:

- Official IIT Jodhpur websites
- Faculty webpage(s)
- Academic regulation PDF documents

Web scraping was performed using `BeautifulSoup`, where all paragraph text was extracted. For PDFs, the `pdfplumber` library was used to extract text from each page.

This resulted in a mixed corpus containing academic, administrative, and research-related text.

3. Data Preprocessing

The collected raw text was cleaned and processed before training the models. The following steps were applied:

- Removed dates, time formats, and special characters
- Converted all text to lowercase
- Tokenized sentences into words
- Removed very short words (length < 3)
- Removed standard English stopwords
- Added custom stopwords such as *"iit"*, *"jodhpur"*, *"institute"* to reduce bias

Finally, the cleaned corpus was saved into a file (`clean_corpus.txt`) for reuse.

4. Corpus Statistics

After preprocessing:

- Total number of documents: 6
- Total tokens: 15441
- Vocabulary size: 3090

These statistics give an idea about the size and diversity of the dataset.

5. Model Training

Two Word2Vec models were trained:

5.1 CBOW Model

- Vector size: 200
- Window size: 8
- Negative sampling: 10
- Trained to predict a word using its context

5.2 Skip-gram Model

- Vector size: 100
- Window size: 5
- Negative sampling: 5
- Trained to predict context words from a target word

Both models were saved for later use.

6. Semantic Analysis

6.1 Word Similarity

We tested words like *student*, *research*, *course*, etc.

The models were able to return contextually related words, showing that meaningful relationships were learned.

Example:

- "student" → related to *course*, *semester*, *registration*
- "research" → related to *project*, *faculty*, *publication*

6.2 Analogy Tasks

Basic analogy operations were performed using vector arithmetic such as:

- course + grade – student

The results were reasonable, showing that the embeddings capture semantic structure.

7. Visualization

To better understand the learned embeddings, we used **t-SNE** to reduce high-dimensional vectors into 2D.

- Words with similar meanings were placed close together
- Academic terms formed clusters (e.g., student-related, course-related)

Separate plots were created for both CBOW and Skip-gram models.

8. Observations

- Skip-gram performed better for capturing semantic relationships
- CBOW was faster and more stable
- Domain-specific stopwords helped improve relevance
- Dataset size affects embedding quality significantly

9. Conclusion

This project demonstrated how word embeddings can be learned from real-world institutional data. Both CBOW and Skip-gram models successfully captured semantic relationships between words. With a larger dataset and better preprocessing, the quality of embeddings can be further improved.